



Anticancer activity of essential oils: Cell cycle perspective

Vanshika Sharma^{a,b}, Deepak Kumar^{c,*}, Kamal Dev^{a,b,d}, Anuradha Sourirajan^{a,b,*}

^a Faculty of Applied Sciences and Biotechnology, Shoolini University, Solan, Himachal Pradesh 173229, India

^b Raj Khosla Center for Cancer Research, Shoolini University, Solan, Himachal Pradesh, India

^c Department of Pharmaceutical Chemistry, School of Pharmaceutical Sciences, Shoolini University, Solan, Himachal Pradesh 173229, India

^d Department of Pharmacology & Toxicology, Wright State University, Dayton, OH 45435-0001, United States of America

ARTICLE INFO

Article History:

Received 1 October 2022

Revised 21 January 2023

Accepted 12 April 2023

Available online 27 April 2023

Edited by Dr. L. McGaw

Keywords:

Medicinal plants

Aromatic plants

Essential oils

Cancer

Cell cycle

ABSTRACT

One of the biggest causes of mortality in humans is cancer, thus emphasizing on research for curing cancer, its prevention and treatment. Mis-regulation of the cell cycle is the underlying cause of any cancer, resulting in uncontrolled cell division leading to tumor formation. Current cancer therapies are limited by side effects, lack of specificity, and toxicity. Essential oils derived from various parts of medicinal and aromatic plants *Citrus limettoides*, *Origanum onites*, *Artemisia campestris*, *Rosmarinus officinalis*, *Thymelaea hirsute*, *Croton matourensis* and flowers and leaves of *Croton micans*, etc., appear to be a promising, safe and effective source of herbal medicine for cancer cure. Essential oils have diverse applications such as aromatherapy, naturopathy, treating various infectious diseases and non-communicable diseases like cancer, as well as for aging. Several essential oils have been reported to possess anticancer activity against cancer cells, as well as in animal models. In this review, we discuss the anti-cancer activity of different essential oils from an array of medicinal plants, and their mechanism of action, with a special focus on the cell cycle targets.

© 2023 SAAB. Published by Elsevier B.V. All rights reserved.

1. Introduction

Cancer is one of the world's major causes of death, yet its effects are not uniformly spread. Due to a multitude of variables, including an aging and expanding population, rapid socioeconomic growth, and changes in the prevalence of risk factors both in developed and developing countries. According to GLOBOCAN 2020, around 19.3 million cases of cancer would be diagnosed worldwide, with roughly 10 million deaths due to cancer (Deo et al., 2022; Cao et al., 2021). Female breast cancer the most common cancer in both sexes in 2020, surpassing pulmonary cancer. Prostate cancer, non-melanoma skin cancer, and colon cancer were placed third, fourth, and fifth, respectively, with stomach, liver, and rectal cancers coming in third, fourth, and fifth. Cervical cancer surpassed oesophageal cancer as the tenth most prevalent type of cancer. In 2020, the top 14 cancer types account for roughly 75% of newly diagnosed cases (69% melanoma of skin cancer was excluded) (Cao et al., 2021).

Around 60% of cancer incidences in Thailand were attributed to cancer of breast, cervix, colorectal, liver, and lungs (Thunb et al., 2020). In India, approximately 67,9421 male cancer patients and 71,2758 female patients were recorded in a study in 2020 (Mathur et al., 2020). Several cancer-causing agents, such as, UV rays, ionizing radiation, and smoke of tobacco, infections and diseases caused by

viruses (such as virus causing hepatitis B triggers cancers i.e., hepatic (liver cancer) and HPV (human papillomavirus) triggers cervical cancer), parasites (schistosomiasis causing bladder cancer), bacteria such as *Helicobacter pylori* (causing stomach cancer), and pollutant edibles by toxins (e.g., hepatic cancer caused by aflatoxin's), certain oxygen-centered radicals can bring a certain type of damage to biomolecules, i.e., oxidative damage such as, deoxyribonucleic acid, lipids (Bhalla et al., 2013). Various drugs are used to treat cancer biomarkers the most important targets in cancer therapy, owing to their specificity. The most commonly used agents for testing and targeting biomarkers include monoclonal antibodies, radiolabelled monoclonal antibodies, antibody-drug conjugates (ADCs), and bispecific T cell engagers (BiTEs) (Kumar et al., 2022; NCI, 2015).

Chemotherapy has a greater effect on cancer cells because they grow and divide quicker than normal cells. Hormone therapy prevent body from producing hormone, or alter the hormone's function which include surgical removal of the gland that produces the hormones (NCI, 2015). Immunotherapy aims at enhancing the population of immune cells and supplement the body with extra nutrients which strengthen the immune system's (NCI, 2015). Radiation treatment can be administered using linear accelerator or radioactive sources implanted in the patient either temporarily or permanently (Kumar et al., 2022; NCI, 2015; Sharma et al., 2021). During cancer surgery, tumor and surrounding tissue are removed. Targeted cancer therapies target specific aspects of cancer cells, such as proteins or

* Corresponding authors.

E-mail addresses: guptadeepak002@gmail.com (D. Kumar), asourirajan@gmail.com (A. Sourirajan).

genes, that aid tumor growth and spread (NCI, 2015 Sharma et al., 2023)

Cancer therapy is associated with a plethora of side effects. Hence complementary and alternative health practices is needed that primarily include the use of natural products, and mind and body practices (National Center for Complementary and Integrative Health). Traditional healing methods and homoeopathy are examples of these approaches (Regassa et al., 2022; Keene et al., 2019). Some of these alternative treatments are proved to be effective such as meditation, yoga, acupuncture, etc. Unconventional approaches in addition to conventional therapy are referred to as 'complementary' therapy, while replacement of conventional therapy is referred to as 'alternative' therapy (Keene et al., 2019). The use of essential oils to treat cancer is an example of alternative therapy. EO's are volatile oils obtained from the leaves, flowers, fruits, and bark of the tree. They are being utilized on the measurable scale for different purposes such as aromatherapies, skin treatment, hair treatment, to repel insects, in various pharmaceutical, cosmetics, and perfume industry. EO's also have antioxidant properties and are being used to treat various diseases that are caused due to disturbed oxidation in biological pathways such as cancer (Halliwell and Gutteridge, 1990; Paz-Elizur et al., 2008), inflammation (Aoi et al., 2008), diabetes (Jain, 2006), Parkinson's disease (Chaturvedi and Beal, 2008), and acquired immunodeficiency syndrome (AIDS) (Sepulveda and Watson, 2002).

1.2. Anticancer activities of essential oils

Essential oils and cancer are trending topics, with researchers trying to implicate essential oils in cancer cure (Patharakorn et al., 2017). Anticancer activity of EO's from various plant species extracted from *Origanum onites* on HepG2 cell line (Özkan and Erdogan, 2011), *Arnica montana* L. on human glioblastoma line T98G and anaplastic astrocytoma MOGGCCM lines (Gawlik-dziki et al., 2020; Sugier et al., 2020), *Rosmarinus officinalis* L. essential oil on various cancer cell lines (Wang et al., 2012). *Labiatae* and *Lauraceae* families have been shown to be cytotoxic against cancer cells (Loizzo et al., 2007), *Cymbopogon flexuosus* on the colon (502,713) and neuroblastoma (IMR-32) cell lines (Sharma et al., 2009). *Morinda citrifolia* EO showed inhibitory activity towards A549 cells (Human lung cancer) (Rajivgandhi et al., 2020). In another study, cytotoxic effect of *Origanum glandulosum* Desf. oil nano emulsions and nano capsules on hepatic cancerous cell line (Hep G-2) (Ali et al., 2020).

Different approaches have been made to treat cancer one of them is targeting the cell cycle. Mutations occur in the genetic control of cell cycles resulting in uncontrolled cell proliferation (A. Kumar et al., 2021). Genes such as pRb and p53 show tumor suppressing activities get deactivated due to mutation and which causes dysfunction in protein regulation in cell progression (Mishra et al., 2017). Cell cycle interruption is linked with cancer follows via mutation of proteins necessary at different levels of the cell cycle (Vermeulen et al., 2003). In a study anti-proliferative essential oils were chosen to examine their influence on the advancement of the cell cycle in budding yeast. Due to the great degree of conservation between yeast and human cell-cycle machinery, yeast research is directly relevant to anticancer drug development (S. Kumar et al., 2021).

2. Anticancer action of essential oils targeting distinct biological pathways

During the cell cycle, different proteins are encoded by different genes; these genes are often referred to as tumor suppressor genes. In order to prevent cancer or tumor growth, these genes are essential for slowing down and mending damaged cells created throughout cell cycles. If a damaged cell cannot be healed, it undergoes apoptosis or cell cycle stop. Cancer may develop from the breakdown of these tumor-suppressing genes. Many biological pathways are employed

to induce cancer cell apoptosis. These pathways and their associated signals are active in normal cells but are suppressed in cancer cells (Gautam et al., 2014; Shukla et al., 2022b). Drugs are used to target the malignant cell by promoting cell death or halting the cell cycle. As a result, compounds extracted from natural sources cause cell death in diseased cells, making them a valuable tool against cancerous cell growth (Gautam et al., 2014). EO's with curative properties might respond in two ways: chemoprevention and cancer suppression. The commencement of detoxifying enzymes, adjustment of DNA repair signaling, anti-metastasis, and anti-angiogenesis are among the several cancer treatment mechanisms. It has been shown that EO's have an effect on the process of tumor inhibition in mammalian replicas, which is one of the many pathways linked with the cytotoxic effects on cancer cells. DNA damage is induced by mutation causes such as UV radiation, carcinogens, and IR rays, which may lead to the formation of tumours or malignancies. Because of this, there are several cell cycles checkpoints that inhibit the reproduction of incomplete or damaged DNA throughout the cell cycle. In order to cure or prevent cancer, some antiproliferative techniques are used to cause cell death in cancer patients. There are two kinds of caspase-dependent processes that lead to cell death: the intrinsic route and the extrinsic pathway (Gautam et al., 2014).

2.1. Intrinsic pathway

The EO's extracted from aerial parts of *Salvia aurea*, *Salvia judaica*, and *Salvia viscosa* and examined for their potential to suppress the growth of human melanoma cell lines M14, A2058, and A375. By stimulating the death of cancer cells via the process of apoptosis, as measured by the Caspase colorimetric test kit, the oils are able to inhibit the progression of the disease (SIGMA RBI St. Louis, USA). This provided scientific evidence for the in vitro usage of these plants in traditional herbal remedies (Russo et al., 2016). The use of real-time PCR was the method that led to the discovery of the effects of *Chenopodium botrys* essential oil on HeLa cells. This method indicate intrinsic apoptotic signals in the mitochondria. After analysing the levels of caspase-3 mRNA expression, researchers came to the conclusion that caspase plays a role in the apoptosis that is caused by *C. botrys* and increased Bax and caspase-3 expression (Fig. 1). The ratio of Bax to Bcl2 rose fivefold when the concentration raised from 50 to 100 g/ml (Rezaieseresht et al., 2020).

2.2. Extrinsic pathway

Significant antioxidant and cytotoxic action were exhibited by the essential oil of *Vitex agnus castus* (VAC EO's) in H69AR cancer cells but not in HEK-293 human normal cells, suggesting a tumor-specific effect. The effect of VAC EO's on Bcl-2, Bcl-XL, Bax, Bad, FADD, Caspase-8, Caspase-9, TRAIL R1/DR4, and TRAIL R2/DR5 on caspase 3/7 activation and apoptosis is described. This analysis demonstrated that VAC EO's are a viable alternative for the creation of new therapeutic drugs for the treatment of multidrug-resistant lung cancer (Fig. 1) (Ilhan, 2020).

2.3. PARP deactivation

PARP poly ADP ribose polymerase is involved in a number of biological processes, including DNA repair, programmed cell death, and genomic integrity. DNA binding domain, auto modification domain, caspase cleaved domain, and complementarity determining domains that are included inside PARP (catalytic domain). PARP is responsible for DNA repair in cancer cells, and inhibiting PARP in cancer cells delays cell death. Histidine and tyrosine residues are found in the catalytic domain, and these are the sites where NAD⁺ interacts to initiate DNA repair. An inhibitor of PARP competes with NAD⁺ and binds to histidine and tyrosine. In this instance, PARP will bind to DNA in

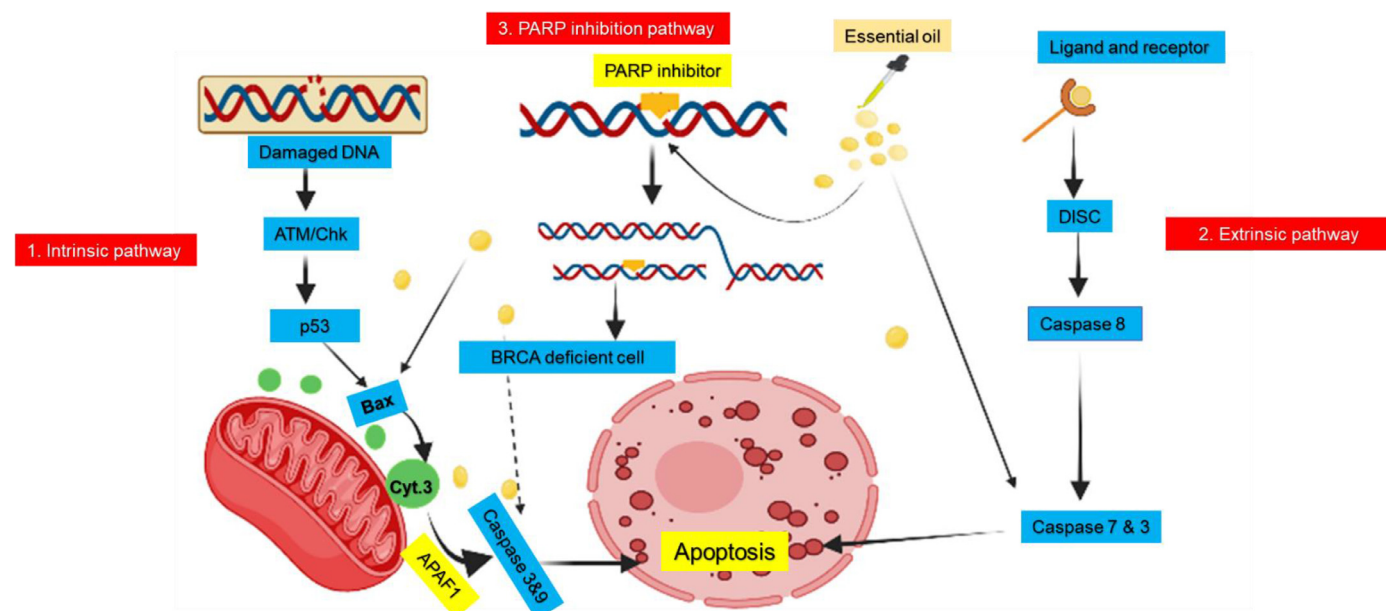


Fig. 1. An overview of effect of essential oils on intrinsic, extrinsic and PARP inhibition pathways operating in cancer. In intrinsic pathway essential oil increases the expression of BAX and caspase 3. For extrinsic pathway essential oil (shown as yellow droplets) acts on caspase 7 and in PARP inhibition pathway, the essential oil prevents the binding of NAD⁺ and catalyzes the binding of PARP inhibitor.

the presence of an inhibitor. The PARP inhibitor halts the DNA replication fork and prevents the synthesis of DNA double-strands. The DNA strand would be defective in BRCA, resulting in poor homologous repair, DSB build up, and ultimate cell death. Both BRCA and PARP domain are the four DNA binding domain, auto modification domain, caspase cleaved domain, and complementarity determining domain are the four domains that are included inside PARP (catalytic

domain). PARP is responsible for DNA repair in cancer cells, and inhibiting PARP in cancer cells delays cell death. Histidine and tyrosine residues are found in the catalytic domain, and these are the sites where NAD⁺ interacts to initiate DNA repair. An inhibitor of PARP competes with NAD⁺ and binds to histidine and tyrosine. In this instance, PARP will bind to DNA in the presence of an inhibitor. The PARP inhibitor halts the DNA replication fork and prevents the

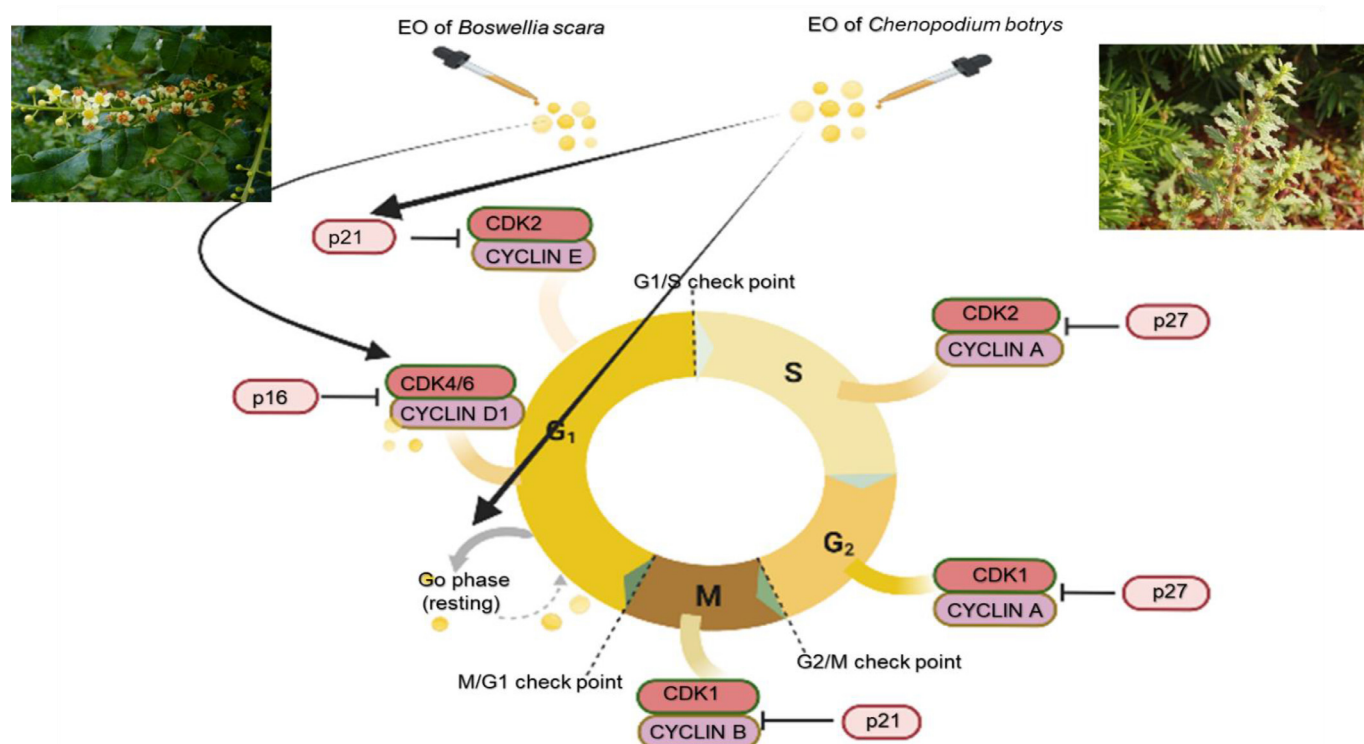


Fig. 2. Anticancer activity of EO's targeting cell cycle. Above figure indicates essential oil, *Boswellia scara* inhibits cdk4 and cyclin D1 and essential oil *Chenopodium botrys* increase the G1/G0 phase of cell cycle and stimulate p21 which was directly navigated by p53, causing an arrest in the cell cycle at the G1 phase.

Table 1

Summary of anticancer activity of various essential oils on cancer cells.

Essential oils	Cancer cells	Mechanism/Activity	Reference
<i>Artemisia campestris</i> and <i>Thymelaea hirsuta</i>	Colon cancer (HT-29) cells	Significant antitumor activity	(Akrouit et al., 2011) (Bisht et al., 2021)
<i>Citrus limettoides</i>	Colon cancer (SW480) cells	Apoptosis via caspase-3 activation and inhibition of cox-2 and IL-6, inflammatory proteins	(Jayaprakasha et al., 2013)
<i>Pulicaria jaubertii</i>	HEPG-2 and MCF-7 human cell lines	Cytotoxicity	(Mag et al., 2013)
<i>Talauma ovata</i> , <i>Symphyopappus itatiayensis</i> , <i>Myrciaria floribunda</i> , <i>Psidium cattleianum</i> , and <i>Nectandra megapotamica</i>	Breast adenocarcinoma (MCF-7), colon adenocarcinoma (KM-12), multiple myeloma (RPMI-8226), prostate carcinoma (PC-3), glioblastoma (SF-268), and lung carcinoma (NCI-H460) cell line	Cytotoxicity	(Souza et al., 2006)
<i>Origanum glandulosum</i> Desf. <i>Chenopodium botrys</i> L.	Hep-G2 (54.93) HeLa cells	Strongest cytotoxic effect Expression of p21 and p53, intrinsic apoptosis pathways and arresting cell proliferation	(Ali et al., 2020) (Rezaieseresht et al., 2020)
<i>Origanum onites</i> <i>Lippia gracilis</i>	HepG2 Mouse melanoma (B16-F10), human hepatocellular carcinoma (HepG2), and chronic myelocytic leukemia (K562) cell lines	Dose-dependent antiproliferative effects G1 arrest in HepG2 cells, apoptosis via remarkable activation of caspase-3.	(Özkan and Erdoğan, 2011) (Ferraz et al., 2013)
<i>Drimys angustifolia</i> and <i>D. brasiliensis</i>	Human bladder carcinoma (T24) and glioblastoma (U-138 MG) cell lines	Apoptosis	(Gomes et al., 2002)
Sage (<i>Salvia aurea</i> L.), (<i>Salvia judaica</i> Boiss.) and (<i>Salvia viscosa</i>)	Human leukemia cell line (HL-60), MCF-7, PC-3, A-549 and MDAMB-231, HL-60 cell.	Inhibits the cell proliferation in and apoptosis by expression of Bcl-xL, caspase-3, -8, -9 and PARP-1 cleavage	(Russo et al., 2016)
<i>Pyrolaea herba</i>	Human chondrosarcoma (SW1353) cells	Increase in the expression of p21, antitumor activity	(Cai et al., 2013)
<i>Curcuma wenyujin</i>	Human hepatoma (HepG2) cell line	Inhibit the growth of HepG2 cells by inducing a cell cycle arrest at S/G2.	(Xiao et al., 2008)
<i>Moroccan thyme</i> <i>Croton matourensis</i> and flowers and leaves of <i>Croton micans</i>	P815 mastocytoma cell line LoVo (colon carcinoma), X-17 (colon carcinoma), HeLa (cervical cancer)	Dose dependent cytotoxic effect Cytotoxicity	(Farmacognosia, 2007) (Compagnone et al., 2010)
<i>Satureja montana</i> L., <i>Coriandrum sativum</i> L. and <i>Ocimum basilicum</i> L.	HeLa, MDA-MB-453, K562 and MRC-5 cell lines	Antiproliferative activity	(Elgndi et al., 2017)
<i>Rosmarinus officinalis</i> L. <i>Myrica gale</i> L.	SK-OV-3, HO-8910 and Bel-7402 human Human lung carcinoma cell line A-549 and human colon adenocarcinoma cell line, DLD-1	Cytotoxic activities Anticancer activities	(Wang et al., 2012) (Sylvestre et al., 2005)
Agarwood essential oil <i>Cedrelopsis grevei</i>	MCF-7 breast cancer MCF-7 cell lines	Anticancer activity Cytotoxicity	(Hashim et al., 2014) (Afoulous et al., 2013)
<i>Pelargonium graveolens</i> (Geranium) and <i>Citrus reticulata</i> (Petitgrain mandarin)	Human promyelocytic leukemia cell lines (HL-60 and NB4)	High anticancer activity	(Fayed, 2009)
Lemon essential oil <i>Rosmarinus officinalis</i>	Human cervical adenocarcinoma cell lines (LeLA) MCF-7 and LNCaP) and one fibroblast cell line (NIH-3T3)	Antiproliferative activity Exhibited antiproliferative activity	(Chunyan et al., 2010) (Hussain et al., 2013)
<i>Sideritis perfoliata</i> , <i>Satureia thymbra</i> , <i>Salvia officinalis</i> , <i>Laurus nobilis</i> and <i>Pistacia palestina</i>	Amelanotic melanoma C32, renal cell adenocarcinoma ACHN, hormone-dependent prostate carcinoma LNCaP, and MCF-7 breast cancer cell line	Cytotoxic activity was found in humulene was tested on the LNCaP cells, but this sesquiterpene was inactive against the MCF-7, C32, and ACHN cell lines	(Loizzo et al., 2007)
<i>Pinus heldreichii</i> Christ (Pinaceae) and <i>P. peuce</i> <i>E. ciliata</i> (Lamiaceae)	HeLa, CaCo-2 and MCF-7 (Human glioblastoma (U87), pancreatic cancer (Panc-1) and triple negative breast cancer (MDA-MB231) <i>in vitro</i>	Cytotoxic effect Low antiproliferative activity	(Basholli-Salih et al., 2023) (Pudziuvelyte et al., 2017)
<i>Achillea fragrantissima</i> (Forssk)	(HCT116): colon cancer cell line and MCF-7 (breast cancer cell line)	Cytotoxic activity	(Choucry, 2017)
Thyme	DLD-1 colorectal cancer cells and L929 fibroblast cells	Antiproliferative and cytotoxic effect to DLD-1 but not to L929 cells	(Temiz et al., 2013)
<i>Annona muricata</i> leaf <i>Coronilla varia</i> <i>Croton flavens</i> L.	Human breast cancer (MCF-7) cell lines MCF7 cancer line Human lung carcinoma cell line A-549 and human colon adenocarcinoma cell line DLD-1	Cytotoxic activity Anticancer activity Cytotoxic activity	(Owolabi et al., 2013) (Dehpour et al., 2014) (Sylvestre et al., 2006)
<i>Houttuynia cordata</i>	HepG2 (hepatocarcinoma), MCF-7 (breast cancer) and NCI-H187 (small cell lung cancer)	Anticancer activity against NCI-H187 cell lines but was non-cytotoxic to HepG2 and MCF-7 human cancer cell lines	(Thunb et al., 2020)
Thymus species from the Pakistani flora <i>Schinus terebinthifolius</i> Raddi <i>Tagetes minuta</i> L. flowers & <i>Ocimum basilicum</i> L. herb	MCF-7, LNCaP and NIH-3T3 cell lines Human breast cancer (MCF-7) Human promyelocytic leukemia cell lines (HL-60 and NB4) and experimental animals model cancer cell line (EACC)	Anti-proliferative activity Anti-proliferative activity Anticancer activity	(A.I. Hussain et al., 2013) (Guzzo da Silva et al., 2019) (Mahmoud, 2013)
<i>Myric rubra</i> leaves <i>Nigella sativa</i> L. <i>Schinus molle</i> L. and <i>Schinus terebinthifolius</i> Radd Myrrh and frankincense	A549 (lung cancer) MCF-7 Human breast cancer cells (MCF-7)	Inhibited cell proliferation Apoptosis Anticancer activities	(Yin et al., 2023) (Periasamy et al., 2016) (Bendaoud et al., 2010)
<i>Arnica montana</i>	MCF-7 and HS-1 Human glioblastoma multiforme T98G and anaplastic astrocytoma MOGGCCM	Myrrh, frankincense and the mixture of essential oils were capable of inducing apoptosis Anticancer activities	(Chen et al., 2013) (Gawlik-dziki et al., 2020; Sugier et al., 2020)

(continued)

Table 1 (Continued)

Essential oils	Cancer cells	Mechanism/Activity	Reference
<i>Micromeria fruticosa</i> (A) and <i>Teucrium Polium</i>	Human breast cancer cell lines (MCF7) and human lung carcinoma cell lines (A549)	Cytotoxic activity	(Al-hamwi et al., 2021)
<i>Eucalyptus camaldulensis</i>	Colorectal cancer cell line Caco-2	Anticancer activity induction of apoptosis	(Taheri et al., 2020)
<i>Nigella sativa</i>	Hep-G2 liver cancer cell line	Anticancer activity	(Gothainayagi and Josephine, 2020)
<i>Ceratophyllum demersum</i>	HT-29 human colon malignant growth cell line	Ethanol extracts were seen as specifically cytotoxic to HT-29	(Awati et al., 2021)
<i>Pistacia palaestina</i> Boiss.	HCT116 Colorectal cancer (CRC)	Anticancer activity	(Awwad et al., 2020)
<i>Eriocephalus africanus</i>	Breast cancer (MCF-7), lung cancer (A549) and liver cancer (HepG2)	Anticancer activity	(Magura et al., 2020)
<i>Euphorbia hirta</i> Linn.	Lung cancer cells NCI-H460 and liver cancer cells HepG2	Inhibitory activity anticancer activities	(Tran et al., 2023)
<i>Morus rotundiloba</i> Koidz.	Human larynx epidermoid carcinoma (Hep2) and human colon adenocarcinoma (SW620) cell lines	Cytotoxicity	(Patharakorn et al., 2017)
<i>Lavandula angustifolia</i> Mill.	A549, H1299, C6 cancer cells and nontumorous HUVEC cells	Antiproliferative activity	(Gezici, 2018)
<i>Juniperus phoenicea</i>	Brain, cervix, lung, liver and breast human cell lines	Cytotoxic activities	(Medicines, 2007)

synthesis of DNA double-strands. The DNA strand would be defective in BRCA, resulting in poor homologous repair, DSB build up, and ultimate cell death. Both BRCA and PARP mechanisms repair DNA in normal cells. In cancer cells, DNA repair is dependent on PARP; hence, in the presence of a PARP inhibitor, cancer cell DNA cannot be repaired, resulting in cell death (Fig. 1). PARP, which is involved in DNA repair in response to environmental stress and is a notable caspase-3 cleavage target, was also cleaved in MDA-MB-231 cells treated with the essential oil of *Boswellia sacra* (Suhail et al., 2011).

3. Cell cycle-targeting anticancer effect of essential oils

3.1. Cell cycle targets

3.1.1. Akt/ERK

The effects of essential oils on the initiation of Akt and ERK1/2 signaling pathways were studied. Essential oil caused a rapid reduction in phospho-Akt (Ser473) levels in each and every breast cancer cell line tested. Cell cycle regulators cdk4 and cyclin D1 both under the influence of the *Boswellia sacra* species control their expression (Fig. 2). Essential oil inhibit the expression of cdk4 in T47D cells, MCF7, and MDA-MB-231 cells, respectively, in all three breast cancer cell lines evaluated throughout time. Cyclin D1 is a vital cell cycle regulator in human breast cancer cells (Suhail et al., 2011).

3.1.2. The effect of essential oil on the G1/G0 phase of the cell cycle

The cell cycle consists of a series of checkpoints and stages, and essential oils attack the cell cycle at a variety of different stages. *Chenopodium botrys* essential oil is made up of monoterpenes and sesquiterpenes, and several studies have shown that sesquiterpenes an essential oil, as natural cancer fighters. Flow cytometry with cells that tagged with PI *Chenopodium botrys* essential oil is able to determine the percentage of cervical cancer cells in each of the four stages of the cell cycle. HeLa cells treated with the essential oil of *C. botrys* for 24 h indicated that the cell cycle persisted in the G1/G0 phase for a longer period of time and that the G2 and S phases were reduced, indicating G1 phase arrest (Fig. 2) (Rezaieseresht et al., 2020).

3.1.3. Targeting p53 and p21

The function of the cyclin-dependent kinase inhibitor (CDKI) protein p21 in the cell cycle is well recognised. Activation of p21 is handled directly by p53, resulting in a cell cycle arrest in the G1 phase (Fig. 2). In the majority of cancer cells, p21 expression is independent of wild-type p53 expression. Increased p21 expression is associated with cell cycle arrest and suppression of metastasis. As a tumor

suppressor protein, p53 is involved in a variety of functions, including halting the cell cycle, repairing damaged DNA and promoting apoptosis, translation and differentiation. To identify the condition of cell growth arrest in HeLa cells the expression levels of the p21 and p53 genes in *C. botrys* essential oil were evaluated. The essential oil of *C. botrys* significantly boosted the expression of p21 and p53 in a concentration-dependent manner, according to their findings (Rezaieseresht et al., 2020; Shukla et al., 2022).

Conclusion and future directions

The present study is an overview of the anti-cancer potential of essential oils that have been shown to target pathways such as PARP inhibition, intrinsic and extrinsic pathways, and the cell cycle to cure cancer. The anti-proliferative action of essential oils has been evaluated using cell lines such as HeLa, MDA-MB-231, and H69AR. These studies show that essential oils target certain pathways like Akt/erk, cell cycle checkpoints and genes in the cell cycle, such as p21, p53, thereby resulting in cell cycle arrest or death. By halting or suppressing malignant cells at a particular phase of the cell cycle, essential oils are excellent candidates for cancer treatments. This research may also be used to combat chemotherapy-related adverse effects such as hair loss, premature aging, etc. Although essential oils have been examined in vitro on a variety of cancer cell lines, there are no studies on their validation in tumor models in vivo. Thus, development of EO's based therapeutics and their cell cycle targets is an attractive area of future research for clinical and pharmaceutical sector.

Table 1.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors would like to thank Raj Khosla centre for Cancer Research, Shoolini University for supporting this project.

References

- Afoulous, S., Ferhout, H., Guy, E., Valentin, A., Moukarzel, B., Couderc, F., Bouajila, J., Toulouse, F., 2013. Chemical composition and anticancer, antiinflammatory,

- antioxidant and antimalarial activities of leaves essential oil of *Cedrelopsis grevei*. Food Chem. Toxicol. 56, 352–362. <https://doi.org/10.1016/j.fct.2013.02.008>.
- Akrout, A., Alarcon, L., El, H., Campa, P., 2011. Antioxidant and antitumor activities of *Artemisia campestris* and *Thymelaea hirsuta* from Southern Tunisia. Food Chem. Toxicol. 49 (2), 342–347. <https://doi.org/10.1016/j.fct.2010.11.003>.
- Al-hamwi, M., Aboul-ela, M., El-lakany, A., & Nasreddine, S. (2021). Growing in Lebanon. 13(1), 103–109.
- Ali, H., Al-Khalifa, A.R., Aouf, A., Boukhebbi, H., Farouk, A., 2020. Effect of nanoencapsulation on volatile constituents, and antioxidant and anticancer activities of Algerian *Origanum glandulosum* Desf. essential oil. Sci. Rep. 10 (1), 1–9. <https://doi.org/10.1038/s41598-020-59686-w>.
- Aoi, W., Naito, Y., Takanami, Y., Ishii, T., Kawai, Y., Akagiri, S., Kato, Y., Osawa, T., Yoshikawa, T., 2008. Astaxanthin improves muscle lipid metabolism in exercise via inhibitory effect of oxidative CPT I modification. Biochem. Biophys. Res. Commun. 366 (4), 892–897. <https://doi.org/10.1016/j.bbrc.2007.12.019>.
- Awati, S.S., Singh, S.K., Wadkar, K.A., 2021. In vitro Antioxidant potential and Anticancer activity of *Ceratophyllum demersum* Linn. extracts on HT-29 human colon cancer cell line. Res. J. Pharm. Technol. 14 (1), 28–36. <https://doi.org/10.5958/0974-360x.2021.00006.8>.
- Awwad, O., Alabbassi, R., Abaza, I.F., Coperchini, F., Rotondi, M., Chiovato, L., Affi, F.U., 2020. Effect of *Pistacia palaestina* Boiss. Essential oil on colorectal cancer cells: inhibition of proliferation and migration. J. Essent. Oil Bear. Plants 23 (1), 26–37. <https://doi.org/10.1080/0972060X.2020.1729244>.
- Basholi-Salih, M., Schuster, R., Hajdari, A., Mulla, D., Viernstein, H., Mustafa, B., & Mueller, M. (n.d.). 2023 Phytochemical composition, anti-inflammatory activity and cytotoxic effects of essential oils from three *Pinus* spp. 10.1080/13880209.2017.1309555
- Bendaoud, H., Romdhane, M., Souchard, J.P., Cazaux, S., Bouajila, J., 2010. Chemical composition and anticancer and antioxidant activities of *Schinus molle* L. and *Schinus terebinthifolius* Raddi berries essential oils. J. Food Sci. 75 (6), 466–472. <https://doi.org/10.1111/j.1750-3841.2010.01711.x>.
- Bhalla, Y., Gupta, V.K., Jaitak, V., 2013. Anticancer activity of essential oils: a review. J. Sci. Food Agric. 93 (15), 3643–3653. <https://doi.org/10.1002/jsfa.6267>.
- Bisht, D., Kumar, D., Kumar, D., Dua, K., Chellappan, D.K., 2021. Phytochemistry and pharmacological activity of the genus *artemisia*. Arch. Pharm. Res. 44, 439–474.
- Cai, L., Ye, H., Li, X., Lin, Y., & Yu, F. (2013). Chemical constituents of volatile oil from *Pyrola* herba and antiproliferative activity against SW1353 human chondrosarcoma cells. 1452–1458. 10.3892/jfo.2013.1816
- Cao, W., Chen, H.Da, Yu, Y.W., Li, N., Chen, W.Q. 2021. Changing profiles of cancer burden worldwide and in China: a secondary analysis of the global cancer statistics 2020. Chin. Med. J. 134 (7), 783–791. <https://doi.org/10.1097/CM9.0000000000001474>.
- Chaturvedi, R.K., Beal, M.F., 2008. PPAR: a therapeutic target in parkinson's disease. J. Neurochem. 106 (2), 506–518. <https://doi.org/10.1111/j.1471-4159.2008.05388.x>.
- Chen, Y., Zhou, C., Ge, Z., Liu, Y., Liu, Y., Feng, W., Li, S., Chen, G., Wei, T., 2013. Composition and potential anticancer activities of essential oils obtained from myrrh and frankincense. Oncol. Lett. 6 (4), 1140–1146. <https://doi.org/10.3892/ol.2013.1520>.
- Choucri, M.A. (2017). Chemical composition and anticancer activity of *Achillea fragrantissima* (Forssk.) Sch. Bip. (Asteraceae) essential oil from Egypt. 9(January), 1–5. 10.5897/JPP2015.0399
- Chunyan, H., Hong, P., Zhenyu, Z., & Jing, S. (2010). Evaluation of antioxidant and antitumor activities of lemon essential oil. 4(18), 1910–1915.
- Compagnone, R.S., Chavez, K., Mateu, E., Orsini, G., Arvelo, F., Suárez, A.I., 2010. Composition and cytotoxic activity of essential oils from *Croton matourensis* and *Croton micans* from Venezuela. Records of Natural Products 4 (2), 101–108.
- Dehpour, A.A., Eslami, B., Rezaie, S., Hashemian, S.F., Shafie, F., Kiaie, M., 2014. Chemical composition of essential oil and in vitro antibacterial and anticancer activity of the hydroalcoholic extract from *Coronilla varia*. Int. J. Biol. Biomol., Agric. Food Biotechnol. Eng. 8 (12), 1414–1417.
- Deo, S.V.S., Sharma, J., Kumar, S., 2022. GLOBOCAN 2020 report on global cancer burden: challenges and opportunities for surgical oncologists. Ann. Surg. Oncol. 29, 6497–6500. <https://doi.org/10.1245/s10434-022-12151-6>.
- Elgndi, M.A., Filip, S., Pavlič, B., Vlačić, J., Stanojković, T., Žizak, Ž., Zeković, Z., 2017. Antioxidative and cytotoxic activity of essential oils and extracts of *Satureja montana* L., *Coriandrum sativum* L. and *Ocimum basilicum* L. obtained by supercritical fluid extraction. Journal of superficial fluids 128, 128–137. <https://doi.org/10.1016/j.sjupflu.2017.05.025>.
- Farmacognosia, R.B.De. (2007). Artigo Chemical composition and antitumor activity of different wild. 17(4), 477–491.
- Fayed, S.A., 2009. Antioxidant and Anticancer Activities of Citrus reticulata (Petitgrain Mandarin) and Pelargonium graveolens (Geranium) Essential Oils. Res. J. Agric. Biol. Sci. 5 (5), 740–747.
- Ferraz, R.P.C., Bomfim, D.S., Carvalho, N.C., Soares, M.B.P., Thanany, B., Machado, W.J., Paula, A., Prata, N., Costa, E.V., Regina, V., Moraes, S., Cesar, P., Nogueira, L., Bezerra, D.P., 2013. Phytomedicine Cytotoxic effect of leaf essential oil of *Lippia gracilis* Schauer (Verbenaceae). Eur. J. Integr. Med. 20 (7), 615–621. <https://doi.org/10.1016/j.phymed.2013.01.015>.
- Gautam, N., Mantha, A.K., Mittal, S., 2014. Essential oils and their constituents as anticancer agents: a mechanistic view. Biomed. Res. Int. <https://doi.org/10.1155/2014/154106>.
- Sugier, P., Jakubowicz-Gil, J., Sugier, D., Kowalski, R., Gawlik-Dziki, U., Kołodziej, B., Dziki, D., 2020. Chemical characteristics and anticancer activity of essential oil from *Arnica montana* L. rhizomes and roots. Molecules 25, 1284. <https://doi.org/10.3390/molecules25061284>.
- Gezici, S., 2018. Original article Promising anticancer activity of lavender (*Lavandula angustifolia* Mill.) essential oil through induction of both apoptosis and necrosis. Ann. Phytomed. 7 (2), 38–45. <https://doi.org/10.21276/ap.2018.7.2.5>.
- Gomes, M.R.F., Schuh, R.S., Jacques, A.L.B., Kelmann, R.G., Koester, L.S., Marina, P., Morrone, F.B., Campos, M.M., Renata, P., 2002. Cytotoxic activity evaluation of essential oils and nanoemulsions of *Drimys angustifolia* and *D. brasiliensis* on human glioblastoma (U-138 lines) in vitro. Revista Brasileira de Farmacognosia 23 (2), 259–267. <https://doi.org/10.1590/S0102-695X2012005000136>.
- Gothainayagi, A., Josephine, N.M., 2020. Characterization of green synthesized gold nanoparticles from *Nigella sativa* oil and evaluation of its antibacterial and anticancer activity. J. Crit. Rev. 7 (12), 1664–1679. <https://doi.org/10.31838/jcr.07.12.272>.
- Guzzo da Silva, B., Foglio, M.A., Vieira e Rosa, P.de T., Taranto, O.P., Frattini Fileti, A.M., 2019. Optimization of hydrodistillation and in vitro anticancer activity of essential oil from *Schinus terebinthifolius* Raddi fruits. Chem. Eng. Commun. 206 (5), 619–629. <https://doi.org/10.1080/00986445.2018.1515074>.
- Halliwell, B., Gutteridge, J.M.C., 1990. The antioxidants of human extracellular fluids. Arch. Biochem. Biophys. 280 (1), 1–8. [https://doi.org/10.1016/0003-9861\(90\)90510-6](https://doi.org/10.1016/0003-9861(90)90510-6).
- Hashim, Y.Z., Phirdaous, A., Azura, A., 2014. Screening of anticancer activity from agarwood essential oil. Pharmacognosy Res. 6 (3), 191–194. <https://doi.org/10.4103/0974-8490.132593>.
- Hussain, A.I., Anwar, F., Chatha, S.A.S., Latif, S., Sherazi, S.T.H., Ahmad, A., Worthington, J., Sarker, S.D., 2013. LWT - Food Science and Technology Chemical composition and bioactivity studies of the essential oils from two *Thymus* species from the Pakistani flora. LWT - Food Sci. Technol. 50 (1), 185–192. <https://doi.org/10.1016/j.lwt.2012.06.003>.
- Ilhan, S., 2020. Essential oils from *Vitex agnus castus* L. leaves induces caspase-dependent apoptosis of human multidrug-resistant lung carcinoma cells through intrinsic and extrinsic pathways. Nutr. Cancer 73 (4), 694–702. <https://doi.org/10.1080/01635581.2020.1823439>.
- Jain, S.K., 2006. Superoxide dismutase overexpression and cellular oxidative damage in diabetes. A commentary on 'Overexpression of mitochondrial superoxide dismutase in mice protects the retina from diabetes-induced oxidative stress. Free Radical Biol. Med. 41 (8), 1187–1190. <https://doi.org/10.1016/j.freeradbiomed.2006.07.017>.
- Jayaprakash, G.K., Murthy, K.N.C., Uckoo, R.M., Patil, B.S., 2013. Chemical composition of volatile oil from *Citrus limetoides* and their inhibition of colon cancer cell proliferation. Ind. Crops Prod. 45, 200–207. <https://doi.org/10.1016/j.indcrop.2012.12.020>.
- Keene, M.R., Heslop, I.M., Sabesan, S.S., Glass, B.D., 2019. Complementary and alternative medicine use in cancer: a systematic review. Complement. Ther. Clin. Pract. 35, 33–47. <https://doi.org/10.1016/j.ctcp.2019.01.004>.
- Kumar, A., Dev, K., Sourirajan, A., 2021. Essential Oils of *Rosmarinus officinalis* L., *Cymbopogon citratus* (DC) Stapf, and the phyto-compounds, delta-carene and alpha-pinene mediate cell cycle arrest at G2/M transition in budding yeast *Saccharomyces cerevisiae*. S. Afr. J. Bot. 141, 296–305. <https://doi.org/10.1016/j.SAJB.2021.05.008>.
- Kumar, S., Sharma, A.K., Lalhlenmawia, H., Kumar, D., 2021. Natural Compounds Targeting Major Signaling Pathways in Lung Cancer. eds In: Dua, K., Löbenberg, R., Malheiros Luzo, A.C., Shukla, S., Satija, S. (Eds.), Targeting Cellular Signalling Pathways in Lung Diseases. Springer, Singapore. https://doi.org/10.1007/978-981-33-6827-9_37.
- Kumar, A., Mirkomil, S., Abbaskhan, T., Shavkatjon, S., Ismatdjan, A., Edwin, M., Abbas, R., Deepak, K., Sadanand, P., 2022. Polymer-based hybrid nanoarchitectures for cancer therapy applications. Polymers (Basel) 14 (15), 3027. <https://doi.org/10.3390/polym14153027>.
- Loizzo, M.R., Tundis, R., Menichini, F., Saab, A.M., Statti, G.A., Menichini, F., 2007. Cytotoxic activity of essential oils from Labiate and Lauraceae families against in vitro human tumor models. Anticancer Res. 27 (5 A), 3293–3299.
- Mag, P., Fawzy, G.A., Ati, H.Y.A.I., & Gamal, A.A.El. (2013). Chemical composition and biological evaluation of essential oils of *Pulicaria jaubertii*. 9(15715). 10.4103/0973-1296.108133
- Magura, J., Moodley, R., Maduray, K., Mackraj, I., 2020. Phytochemical constituents and in vitro anticancer screening of isolated compounds from *Eriocapellus africanus*. Nat. Prod. Res. 0 (0), 1–4. <https://doi.org/10.1080/14786419.2020.1744138>.
- Mahmoud, G.I., 2013. Biological effects, antioxidant and anticancer activities of Marigold and Basil essential oils. J. Med. Plants Res. 7 (10), 561–572. <https://doi.org/10.5897/JMPR12.350>.
- Mathur, P., Sathishkumar, K., Chaturvedi, M., Das, P., Sudarshan, K.L., Santhappan, S., Nallasamy, V., John, A., Narasimhan, S., Roselind, F.S., 2020. Cancer statistics, 2020: report from national cancer registry programme, India. JCO Global Oncol. 6, 1063–1075. <https://doi.org/10.1200/go.20.00122>.
- Medicines, A., 2007. of Essential Oils of Leaves and Berries of *Juniperus phoenicea* L. Grown 4 (4), 417–426.
- Mishra, R., Sareen, S., Sharma, B., Goyal, S., Kaur, G., Bhardwaj, S., Siddiqui, A.A., Husain, A., Singla, R.K., Rashid, M., et al., 2017. Phenothiazines and related drugs as multi drug resistance reversal agents in cancer chemotherapy mediated by p-glycoprotein. Curr. Cancer Ther. Rev. 13, 28–42. <https://doi.org/10.2174/1573394713666170524122904>.
- NCI. (2015). Types of Cancer Treatment - National Cancer Institute. In National Cancer Institute (p. 368). <https://www.cancer.gov/about-cancer/treatment/types>
- Owolabi, M.S., Ogundajo, A.L., Dosoky, N.S., Setzer, W.N., Annona, K., Composition, E.O., 2013. The cytotoxic activity of *Annona muricata* leaf oil from Badagary. Nigeria 1 (1), 1–3.
- Özkan, A., Erdoğan, A., 2011. A comparative evaluation of antioxidant and anticancer activity of essential oil from *Origanum onites* (Lamiaceae) and its two major phenolic components. Turkish J. Biol. 35 (6), 735–742. <https://doi.org/10.3906/biy-1011-170>.
- Patharakorn, T., Arpornsuwan, T., Wetprasit, N., Promboon, A., & Ratanapo, S. (2017). Antibacterial activity and cytotoxicity of the leaf essential oil of *Morus rotundifolia* Koidz. 4(9), 837–843. 10.5897/JMPR10.131

- Paz-Elizur, T., Sevilay, Z., Leitner-Dagan, Y., Elinger, D., Roisman, L.C., Livneh, Z., 2008. DNA repair of oxidative DNA damage in human carcinogenesis: potential application for cancer risk assessment and prevention. *Cancer Lett.* 266 (1), 60–72. <https://doi.org/10.1016/j.canlet.2008.02.032>.
- Periasamy, V.S., Athinarayanan, J., Alshatwi, A.A., 2016. Anticancer activity of an ultrasonic nanoemulsion formulation of *Nigella sativa* L. essential oil on human breast cancer cells. *Ultrason. Sonochem.* 31, 449–455. <https://doi.org/10.1016/j.ultsonch.2016.01.035>.
- Pudziulevityte, L., Stankevicius, M., Maruska, A., Petrikaite, V., Ragazinskiene, O., Draksiene, G., Bernatoniene, J., 2017. Chemical composition and anticancer activity of *Elsholtzia ciliata* essential oils and extracts prepared by different methods. *Ind. Crops Prod.* 107 (May), 90–96. <https://doi.org/10.1016/j.indcrop.2017.05.040>.
- Rajivgandhi, G., Saravanan, K., Ramachandran, G., Li, J.L., Yin, L., Quero, F., Alharbi, N.S., Kadaikunnan, S., Khaled, J.M., Manoharan, N., Li, W.J., 2020. Enhanced anti-cancer activity of chitosan loaded *Morinda citrifolia* essential oil against A549 human lung cancer cells. *Int. J. Biol. Macromol.* 164, 4010–4021. <https://doi.org/10.1016/j.ijbiomac.2020.08.169>.
- Regassa, H., Sourirajan, A., Kumar, V., Pandey, S., Kumar, D., Dev, K., 2022. A Review of Medicinal Plants of the Himalayas with Anti-Proliferative Activity for the Treatment of Various Cancers. *Cancers (Basel)* 14 (16), 3898. <https://doi.org/10.3390/cancers14163898>.
- Rezaieseresht, H., Shobeiri, S., Kaskani, A., & Article, O. (2020). *Chenopodium Botrys* Essential Oil as A Source of Sesquiterpenes to Induce Apoptosis and G1 Cell Cycle Arrest in Cervical Cancer Cells. 19(October 2018), 341–351. 10.22037/ijpr.2019.1100671
- Russo, A., Formisano, C., Rigano, D., Cardile, V., Arnold, N.A., Senatore, F., 2016. Comparative phytochemical profile and antiproliferative activity on human melanoma cells of essential oils of three Lebanese *Salvia* species. *Ind. Crops Prod.* 83, 492–499. <https://doi.org/10.1016/j.indcrop.2015.12.080>.
- Sepulveda, R.T., Watson, R.R., 2002. Treatment of antioxidant deficiencies in AIDS patients. *Nutr. Res.* 22 (1–2), 27–37. [https://doi.org/10.1016/S0271-5317\(01\)00355-4](https://doi.org/10.1016/S0271-5317(01)00355-4).
- Sharma, P.R., Mondhe, D.M., Muthiah, S., Pal, H.C., Shahi, A.K., Saxena, A.K., Qazi, G.N., 2009. Anticancer activity of an essential oil from *Cymbopogon flexuosus*. *Chem. Biol. Interact.* 179 (2–3), 160–168. <https://doi.org/10.1016/j.cbi.2008.12.004>.
- Sharma, A., Shambhwani, D., Pandey, S., Singh, J., Lahlhenmawia, H., Kumarasamy, M., Singh, S.K., Chellappan, D.K., Gupta, G., Prasher, P., Dua, K., Kumar, D., 2023. Advances in Lung Cancer Treatment Using Nanomedicines. *ACS Omega* 10 (41), 1. <https://doi.org/10.1021/acsomega.2c04078>.
- Shukla, A., Thakur, M., Verma, R., Lahlhenmawia, H., Bhattacharyya, S., Bisht, D., Singh, A., Parcha, V., Kumar, D., 2022. Unravelling the therapeutic potential of orchid plant against cancer. *S. Afr. J. Bot.* 150, 69–79.
- Shukla, M.K., Dubey, A., Pandey, S., Singh, S.K., Gupta, G., Prasher, P., Chellappan, D.K., Oliver, B.G., Kumar, D., Dua, K., 2022. Managing Apoptosis in Lung Diseases using Nano-assisted Drug Delivery System. *Curr. Pharm. Des.* 28 (39), 3202–3211. <https://doi.org/10.2174/1381612828666220413103831>.
- Suhail, M.M., Wu, W., Cao, A., Mondalek, F.G., Fung, K.-M., Shih, P.-T., Fang, Y.-T., Woolley, C., Young, G., & Lin, H.-K. (2011). *Boswellia sacra* essential oil induces tumor cell-specific apoptosis and suppresses tumor aggressiveness in cultured human breast cancer cells. 10.1186/1472-6882-11-129
- Sylvestre, M., Legault, J., Dufour, D., Pichette, A., 2005. Chemical composition and anticancer activity of leaf essential oil of *Myrica gale* L. *Phytomedicine* 12 (4), 299–304. <https://doi.org/10.1016/j.phymed.2003.12.004>.
- Sylvestre, M., Pichette, A., Longtin, A., Nagau, F., Legault, J., 2006. Essential oil analysis and anticancer activity of leaf essential oil of *Croton flavens* L. from Guadeloupe. *J. Ethnopharmacol.* 103 (1), 99–102. <https://doi.org/10.1016/j.jep.2005.07.011>.
- Taheri, E., Ghorbani, S., Safi, M., Sani, N.S., Amoodizaj, F.F., Heidari, M., Chavoshi, R., Hajazimian, S., Isazadeh, A., Heidari, M., 2020. Inhibition of colorectal cancer cell line Caco-2 by essential oil of *Eucalyptus camaldulensis* through induction of apoptosis. *Acta Med. Iran.* 58 (6), 260–265. <https://doi.org/10.18502/acta.v58i6.4051>.
- Temiz, T., Altun, A., & Kaya, T. (2013). Thyme essential oil inhibits proliferation of DLD-1 colorectal cancer cells through antioxidant effect. 14–24.
- Thunb, H., Khamsan, S., Intecha, N., & Inpeng, P. (2020). Chemical Compositions and Anticancer Activity of Essential Oil from Department of Chemistry, School of Science, University of Phayao, Demonstration School, University of Phayao, Phayao, 56000 Department of Environment Science, School of Energy and. 17(April), 23–31.
- Tran, N., Nguyen, M., Le, K.P., Nguyen, N., Tran, Q., Le, L., 2023. Screening of antibacterial activity, antioxidant activity, and anticancer activity of *Euphorbia hirta* Linn. Extracts. <https://doi.org/10.3390/app10238408>.
- Vermeulen, K., Van Bockstaele, D.R., Berneman, Z.N., 2003. The cell cycle: a review of regulation, deregulation and therapeutic targets in cancer. *Cell Prolif.* 36 (3), 131–149. <https://doi.org/10.1046/j.1365-2184.2003.00266.x>.
- Wang, W., Li, N., Luo, M., Zu, Y., Efferth, T., 2012. Antibacterial activity and anticancer activity of *Rosmarinus officinalis* L. essential oil compared to that of its main components. *Molecules* 17 (3), 2704–2713. <https://doi.org/10.3390/molecules17032704>.
- Xiao, Y., Yang, F., Li, S., Hu, G., Lee, S.M., & Wang, Y. (2008). Essential oil of *Curcuma wenyujin* induces apoptosis in human hepatoma cells. 14(27), 4309–4318. 10.3748/wjg.14.4309
- Yin, J., Li, X., Huang, F.F., Lu, M.H., Yang, J., Zhu, L.Y., & Zhu, Y. (n.d.) 2023. IOP Conference Series: earth and Environmental Science Chemical composition, antioxidant and anticancer activity of the essential oil from *Myric rubra* leaves 10.1088/1755-1315/346/1/012085
- Sharma, L., Sharma, A., Nandy, S. K., Kumar, A. P., Singh, S., Chellappan, D. K., Gupta, G., Dua, K., Kumar, D., 2021. Radiotherapy in covid-19: a review. *Pharmacologyonline* 2, 277–285.